ISSN 1831-9424



Ontology Learning for Hybrid Threats

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2025



Joint
Research
Centre

EUR 40199

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JRC140863

EUR 40199

PDF ISBN 978-92-68-23985-8 ISSN 1831-9424 doi:10.2760/8060534 KJ-01-25-042-EN-N

Luxembourg: Publications Office of the European Union, 2025

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How to cite this report: European Commission: Joint Research Centre, Bosso, F. and Ruberto, S., Ontology Learning for Hybrid Threats, Publications Office of the European Union, Luxembourg, 2025,
https://data.europa.eu/doi/10.2760/8060534, JRC140863.

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1 Abstract

This technical report discusses the potential use of advanced Artificial Intelligence technologies for mining Open Source Information in the context of Hybrid Threat analysis. Increasing international tensions depict a scenario where Hybrid Threats acquire even more importance than in past years and monitoring such activities poses important technological challenges. In this scenario, Large Language Models show promising results but also the corresponding fundamental limitations. Hallucinations, biases, and lack of knowledge are emerging as issues difficult to tackle, especially in the field of Hybrid Threat analysis. We propose the use of formal knowledge bases to correct Large Language Model issues. Despite their interesting applications knowledge bases require human-intensive labor. After exploring the state-of-the-art we propose a methodology to reduce the human effort and learn automatically part of the Knowledge base. This last step involves again Large Language Models but, with the precaution of exploiting only a simple restricted set of linguistic abilities, we avoid the common pitfalls observed before. We expect to enhance the quality of Open Source Intelligence by having more reliable analysis of textual data.

2 Introduction

We Introduce the concepts of Hybrid Threats, Large Language Models, and Knowledge Graphs that represent the starting points of this work. After a general description, we highlight the open issues that this work is addressing.

2.1 Hybrid threats

The European Union and its neighborhood face a growing threat from hybrid attacks, which combine military and non-military tactics to achieve strategic objectives. These threats encompass cyberattacks, damage to critical infrastructure, disinformation campaigns, radicalization of politics, and the exploitation of migration(HTd, 2018). They pose a significant danger to European security and stability, specifically targeting democratic systems and countries in the process of modernization. The EU is adopting a comprehensive approach to address emerging security threats, integrating internal, external, digital, and physical security. This is reflected in two key policy initiatives: the EU Security Union Strategy of 2020 (European Commission, 2020), which aims to boost EU security and stability, and the Strategic Compass for Security and Defence of 2022 (European Council, 2022), which outlines a holistic approach to security, combining digital and physical aspects. The EU is prioritizing a unified and multidimensional security strategy to address complex threats effectively. Following the policymakers, the Joint Research Centre of the European Commission (JRC), in partnership with the Helsinki-based European Centre of Excellence for Countering Hybrid Threat (Hybrid-CoE), presented a conceptual model that captures the complexity of HTs offering a powerful tool to understand and communicate them. This Conceptual Model (Giannopoulos, G. et al., 2021) introduces an analytical framework that defines HTs' building blocks like *Actors* (and their *strategic objectives*), *Tools* applied by the actors, *Domains* that are targeted, and *Phases* (including the types of *activity* observed in each phase. The conceptual model describes how Hostile actors, including state and non-state entities, employ a combination of tools to achieve strategic objectives, exploiting vulnerabilities and opportunities across multiple domains. These tools can have direct or cascading effects, targeting one or multiple domains, and can be used to create new vulnerabilities or take advantage of existing ones. The ultimate goal is to impact a specific domain, which may not be the same as the one where the initial activity is detected, highlighting the complexity and unpredictability of hybrid threats. Understanding HTs was just the first step. In the effort to enhance societal resilience against HT, JRC and the Hybrid-CoE published "The Comprehensive Resilience Ecosystem (CORE) model (Jungwirth R et al., 2023) that serves as a cornerstone in this field by providing a framework for analysing and responding to HT activities. The CORE model "combines resilience and hybrid threats in one new systemic approach. To increase resilience to hybrid threats, the EU must be understood as a comprehensive yet complex system in which multiple interconnected adaptive systems (spaces, layers, and domains) interact synchronously"(Jungwirth R et al., 2023).
Many of the previously mentioned HT activities are based on natural language (e.g. social network posts, news, advertisements, etc.) or leave traces in some form of online texts (declarations, news, etc.). The ability to analyze substantial amounts of text is therefore crucial, especially if we aim at monitoring HT activities with an Open Source Intelligence (OSINT) paradigm. OSINT is a practice referring to the search and extraction, from openly available data sources, of useful information to address specific questions and produce actionable intelligence. This information can then be used to support different sectors ranging from national security to law enforcement (European Commission, 2022). Artificial Intelligence (AI) breakthrough promises to extend the capabilities of Natural Language Processing (NLP) to an unprecedented level. The major advances in this field have been promoted by the wide spreading of the so-called Large Language Models (LLM). Although powerful and able to process vast amounts of text, they suffer some major limitations. This work aims to review some of these limitations and propose a method to extract meaningful information from raw text by producing outputs that can be saved in well-organized data structures for different downstream tasks related to HT monitoring.

2.2 Large Language Models and Knowledge Graphs

A large language model represents the linguistic application of a Deep Learning (DL) architecture known as transformers. This transformer architecture, introduced by (Vaswani et al., 2023) in 2017 to enhance the accuracy of machine translation tasks, is now extensively utilized across many AI subfields. One common application is in Computer Vision, where Vision Transformers (ViT) have been widely adopted for tasks such as object detection and tracking (Li et al., 2022) (Chen et al., 2021). Transformers have revolutionized NLP due to

their capability to generate human language by leveraging deep neural networks with many parameters. These models are trained on extensive amounts of linguistic data, enabling them to perform a variety of NLP tasks—including text generation, translation, summarization, and question-answering—while maintaining coherence.

A large language model typically employs mechanisms such as self-attention to capture contextual relationships among words in a sequence. The scale of these models, which can reach billions or even trillions of parameters, combined with the vast amount of data used for training, allows the transformer architecture to generalize effectively across different domains and generate coherent, well-structured human-like text. However, many of these models, especially those with a high number of parameters, are characterized by a highly computationally demanding inference phase. Consequently, big models require High Performance Computing (HPC) platforms equipped with specialized hardware (e.g., GPU, TPU, LPU) to operate effectively.

More recently, knowledge graphs (KG) have been widely adopted and have proven to be fundamental in enhancing the performance of LLMs. Using KG data structure in the LLM context simplifies the procedure of providing coherent background knowledge to the inference process, giving some assurance about LLM answers' quality (Yang et al., 2024, Agrawal et al., 2023). Simplifying, when the LLM does not have enough data for a particular situation, the training process was not successful, or the prompting is not effective, the answers by the LLM may not be accurate or, even worse, not logical or true at all. In these cases, we say that LLM is hallucinating. By comparing LLM answers to well-structured knowledge graph background information it is possible to check coherence and truthfulness of the output and try to correct it. KG can be used also as input in multiple steps inference processes limiting the hallucination risk even further (Agrawal et al., 2023). Indeed, KGs allow for the provision of information to LLMs, thereby improving their ability to understand the input context (Liu et al., 2019). This has been shown to reduce open issues associated with LLMs, such as hallucination (Baek et al., 2023), and to broaden their domain of knowledge without the necessity of fine-tuning on up-to-date data. For this reason, KGs can play a crucial role in leveraging LLMs for meaningful information extraction from open sources' raw text.

A KG can be defined as a non-linear type of data structure composed of entities, such as events and objects, and the relationships between them. KGs are thus made up of two essential components: nodes and edges. Ontologies are often discussed in the context of KGs, although it is still debated whether the two concepts coincide. For the purposes of this work Ontologies and Knowledge Graphs are equivalent, we will use KG later in the text because it is a more general term, not necessarily bounded to a specific approach.

Ontologies primarily impose two requirements for definition: (i) formal and explicit specifications of the concepts within a domain, and (ii) relationships between those concepts. Additionally, they can encompass rules governing both concepts and relationships. An example of a rule from (Verhodubs, 2014) is as follows: Let the ontology consist of the concepts *Vehicle*, *Car*, and *Auto*, with the relationships *part_of*—from the concept *Car* to the concept *Vehicle* to indicate that a car is a type of vehicle—and the relationship *equivalent*—linking the concepts *Car* and *Auto* to denote that they are synonymous within this ontology (Figure 2.1).

Given this structure it is possible to generate a rule as

IF *Car* **EQUIVALENT** *Auto* **THEN** (*part_of* **Vehicle**) $\in$ *Auto.*

One fundamental property of ontologies is their ability to store knowledge in a structured manner that can be easily parsed and manipulated by computers. Ontologies also facilitate the process of knowledge sharing and reuse (Wong et al., 2011, Gruber, 1993).

Currently, ontologies have been effectively used across a diverse range of tasks. They play a crucial role in information extraction by assisting in identifying relevant data from vast amounts of unstructured information and organizing it into a well-defined and formal structure (Buitelaar et al., 2008, Saggion et al., 2007). Within the scientific community, ontologies are a key step in standardizing definitions, ensuring that researchers from different disciplines have a common ground truth for terms and concepts (Stevens and Lord, 2009).

Ontologies are not a given oracle, to exploit an ontology in whatever field it needs to be created first. This job can be either done by field-experts via manual construction, defining a hierarchy of fundamental concepts and the respective relationships, or can be automated by directly learning the ontology. Even if the second is an appealing possibility under the operational point of view, fully automated ways of learning ontology has been shown to be challenging (Wong et al., 2011, Du et al., 2024). The recent advancements in AI and the capabilities of LLMs to process complex texts might represent a shift of paradigm and open a set of new possibilities. In literature, this paradigm of automatic ontology creation by means of AI is mentioned as Ontology Learning (OL). Please note that LLMs play a different role in the OL field and in the OSINT domain. The key idea is that for the OL tasks LLM can answer simpler queries in a finite context so exploiting the LLM

Figure 2.1: Ontology schematics comprising concepts and relationships

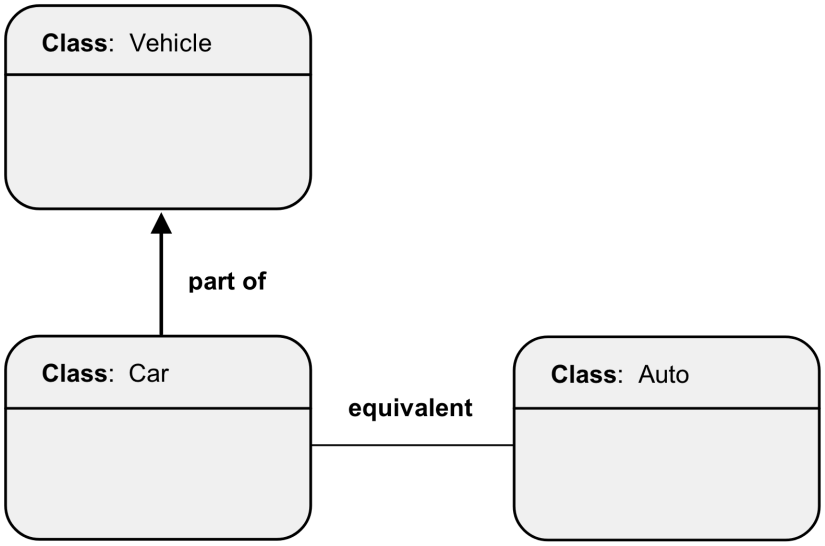
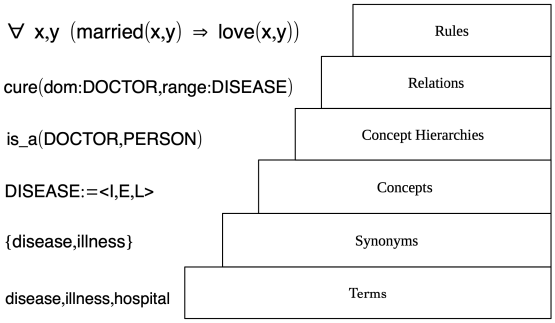


Figure 2.2: Schematics of the Ontology Learning Layer Cake framework



Source: (Wong et al., 2011)

potential having less risk of low-quality answers. For example, during OL process LLM output can be checked for consistency at each step against the background knowledge already in the KGs. In this case the verification process can exploit logic tools already developed for reasoning on KGs, giving the chance to early detect hallucinations, contradictions, biases, insufficient domain knowledge, obsolete knowledge, etc. in LLMs output.
A prominent framework for ontology learning categorizes the process into six distinct layers depicted in Figure 2.3: terms, synonyms, concepts, hierarchies, relations, and rules. This approach is often referred to as the "ontology learning layer cake" a concept popularized by Buitelaar et al. (Buitelaar et al., 2005). This is a fundamental framework for systematically organizing and developing ontologies, as it ensures that all needed components are considered and that the ontology resulting from the process is comprehensive and well-structured. By breaking down the ontology learning process into these layers, it is possible to effectively tackle the complexities related to the construction of robust ontological models.
At the moment, the manual construction is the predominant way of creating ontologies and it remains a labor-intensive task.
Because of that, the research and development of reliable methods for automating the process of ontology creation has remained of great interest. Most traditional OL approaches have been focusing on techniques involving statistics, linguistics, and logic (Wong et al., 2011, Asim et al., 2018, Patel and Debnath, 2024). Now, with the latest advancement in NLP induced by the introduction of LLMs, a paradigm shift is emerging. This shift is growing on the ability of LLMs of performing a wide range of different NLP tasks (Yang et al., 2023). Based on that, the research question that is emerging concerns the potential ability of these models to perform OL tasks.
Current research efforts regarding the exploitation of LLMs for OL are progressing rapidly. Given their capability

to perform general-purpose NLP tasks, such models have proven valuable in assisting ontologists during the manual creation of ontologies. Several tools have been developed on or around platforms like ChatGPT to generate competency questions for ontology development (Zhang et al., 2024). Another significant area of research focuses on converting natural language sentences into OWL axioms to enrich ontologies (Mateiu and Groza, 2023). These research efforts highlight how LLMs can effectively reduce the effort associated with some of the most challenging aspects of ontology creation, making them well-suited for ontology-related tasks.
Semi-automated strategies for ontology creation leveraging LLMs have been proposed. Kommineni et al. (Kommineni et al., 2024), for instance, describe an approach that translates the traditional manual methods for ontology creation into a set of prompts passed as input to ChatGPT. In the first stage, the dataset is fed to ChatGPT to create the so-called competency questions (CQs). Secondly, ChatGPT is used for the extraction of relations and concepts exploiting these CQs. As the third and last step, the obtained information is formalized into an ontology. This study is a tangible result to highlight the ability of LLMs in reducing the human effort in the process of ontology creation. Still, a human-in-the-loop approach aimed at evaluating and adjusting the produced output is needed.
Another example is the work by Funk et al. (Funk et al., 2023). They introduced a method for automatic ontologies construction tailored to a specific domain. Unlike OL layer cake, this method involves an algorithm that iteratively queries GPT-3.5-turbo. The iterative prompting is aimed at generating an ontology based on an initial seed concept defining the domain. Different experiments show that the framework can perform an automatic ontology construction but still require a domain expert to ensure quality and consistency of the ontology.
Research is now concerned with investigating the right amount of ontology simplification such that they can be constructed correctly in an automatic manner. This is because even simplified ontologies can be valuable for specific domains as they still provide a rigorous and meaningful way to store domain-related data.
To understand why the integration of KGs and LLMs could be beneficial for an NLP system, we recall here that LLMs are known to show relevant issues with hallucinations and factual inaccuracies (Rawte et al., 2023, Ji et al., 2023).
The origin of this problem concerns the heavy dependence of LLMs on patterns contained in the training data (Ray, 2023). This is especially worrying in a security context like HT monitoring. To solve these issues, ontologies and KG have been proposed as potential solutions to enhance LLMs performance and reliability by reducing hallucinations (Agrawal et al., 2023, Tonmoy et al., 2024).
For instance, incorporating structured knowledge into LLMs prompts may guide their responses towards more accurate and reliable information (Jiang et al., 2023). Additionally, exploiting knowledge graphs and the related knowledge bases (KB) (i.e.: the data populating the KG) for validation of LLMs output, ensures that the generated data is cross-checked against verified data, thereby reducing the likelihood of inaccuracies (RobertL.Logan et al., 2019) (e.g.: Retrieval Augmented Generation (RAG) paradigm).
Bounding the answer to a KG lowers flexibility that LLMs alone could have but provides greater guarantees in terms of correctness. Moreover, such a KG can serve as a reliable base for several downstream tasks requiring high accuracy and quantification based on ground truth data. One practical example, especially for this work, is the analysis and monitoring of HT activities where the computation of statistics, metrics and the accurate, bias-free recognition of patterns is critical. Alternatively, KGs can be provided during the pre-training, training, and fine-tuning phases of LLM development. In this case the model could result in learning more robust and accurate information representations (Baldazzi et al., 2023). In this work we do not explore these last possibilities because they all require some amount of model training that is expensive and require specialized hardware and personnel.
Overall, by reducing brittleness phenomena of LLMs on one side, and providing a solid base for quantitative analysis on the other, KGs can act as fundamental keystones by enhancing both reliability and (supervised) applicability of AI systems in data-driven decision-making process.

3 Framework proposal

3.1 HYbrid-threat Based Ontology Learning Tool - HYBOLT

The findings produced by the present work provide a partial contribution to the creation of a system able to transform unstructured text into structured data representation. The HYbrid-threat Based Ontology Learning Tool (HYBOLT) is a framework tailored on the HT domain that aims at inferring a domain-related KG from the provided text and populate it with reliable data. We will illustrate how the different concepts introduced in Section 2 contributes to the implementation of the HYBOLT proposal. In case of hallucination, for instance, having a KG populated with a set of reliable data allows developers to create logical mechanisms to mitigate the generation of false information as much as possible. HYBOLT takes as input a text describing a terrorist event and can extract, for instance, where it happened similarly to what has been described in (Bosso et al., 2023). Assuming that we are interested in high-level areas rather than specific town names, having a KG filled with hierarchically-related geographical data (e.g. : Ispra *is in* Varese *is in* Lombardia *is in* Italy) allows to accurately up-scale location information drastically reducing the chance of errors that may be derived by recursively query LLMs.
From a higher-level perspective, the long-term goal is to conceptualize and build a framework that can be wrapped around any LLM, and which provides it with two main pipelines as in Figure 3.1. The first component is the *ingestion pipeline* that is responsible for importing open information coming from different sources. This pipeline converts unstructured text into a knowledge graph, taking care of building entities and relationships among them. Ideally this process will enforce a certain level of coherence in the KG giving a uniform description for the events and, depending on the application preferences, signalling contradictions and gaps.
The second component is the *querying pipeline* that allows the end user to query the knowledge formalized by the previous pipeline by means of different tools. Specifically, tools of the querying pipeline can be intended as pieces of code previously developed to process the output of the LLM or to assist the language model itself in performing the requested task. They are pre-implemented routines that the LLM can exploit to expand his domain of action with the aim of fulfilling the prompted task. For instance, this can be done by writing routines to exploit pre-existing and verified KG linked to the system to produce source-informed answers; Another example could be the dynamic addition of pre-implemented prompt snippet to the user input in order to exploit contextual LLM behavior (e.g.: Act as a Hybrid Threat expert, etc.). An additional example could be a vector database that might be used to keep track of important KG uploaded by the user (declarative memory) and/or previous interaction user-LLM (episodic memory), allowing the implementation of the RAG paradigm.
Some of the advantages that the HYBOLT framework can gain from the interaction of the two previously introduced pipelines are: (i) expand the knowledge domain of the LLM with information that were not available before its training phase; (ii) provide the model with domain specific knowledge without having to fine-tune it; (iii) reduce hallucination, and in general, the brittleness of the system.
We want the final system to ingest a set of openly available text data and transform it into a structured data representation by either creating a KG from scratch or updating an existing one. Once the KG is available, it can be exported and used by analysts for different tasks such as statistical analysis and pattern recognition. On top of that, we aim at implementing an interface, connected to the LLM engine, through which the user can ask for the execution of pre-implemented tools for quick analysis.
The HYBOLT framework is part of a vision that we aim to create by developing and assembling all the blocks depicted in Figure 3.1. In the next section, we introduce the block we focused on for the current research activity: the triplet extractor.

3.2 From text to Knowledge Graphs

A KG is formally defined as a collection of atomic data entities, representing the smallest units of information that the KG can capture. These fundamental units collectively enable the reconstruction of the entire Knowledge Graph. Among the most used conventions to represent KG we have the Resource Description Framework (RDF) where the atomic data entity takes the name of *triplet* (W3C, 2024). The minimal graph model represents a predicate that is true for a subject and an object. The resulting minimal KG is represented in figure 3.3.
To build a KG useful to produce actionable intelligence, the first step is the triplets extraction. This process is supported by the language model.

Figure 3.1: High-level schematic of the HYBOLT framework. At the top is an ingestion pipeline that aims to transform unstructured text into structured data representation. In the bottom part, a querying pipeline aimed at producing source informed answer based on the collected and structured knowledge

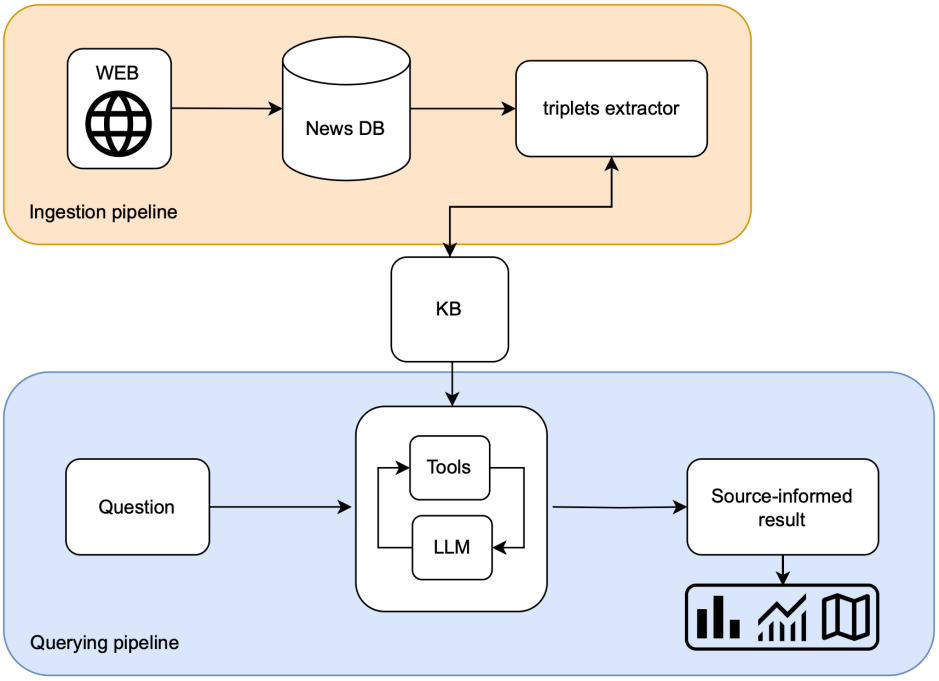
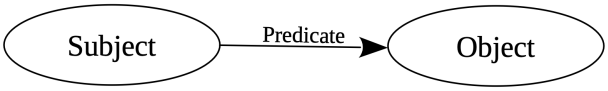


Figure 3.2: An RDF graph with two nodes (Subject and Object) and a triple connecting them (Predicate).



Source: (W3C, 2024)

As mentioned previously, the LLM models might suffer from phenomena such as hallucinations and produce outputs that are not reliable. Our aim here is to create a triplet extractor attenuating this issue as much as possible to produce reliable KGs.

With this work, we wanted to investigate the effectiveness of an approach to reduce hallucination consisting in exploiting only basic LLMs abilities to extract KG from snippet of text. The key point here is that by using only the most basic level of analysis, the closest possible to a grammatical decomposition, we do not rely on field specific knowledge or complex logical reasoning. With this approach we try to avoid biases, hallucinations, and lack of background knowledge since only basic linguistic capabilities are required from the LLM.

To develop this work, we decided to use *nous-hermes-2-mixtral-8x7b-dpo* LLM model. The reason behind this choice is twofold. On one side, using a model with fewer parameters potentially allows the final system to run locally, without the need to outsource computational capabilities during the inference phase. This accounts also for energy savings policy that are addressing the worrying increase of resource consumption related to AI and LLM. On the other side, being able to achieve meaningful results with smaller models leaves us with a model complexity buffer exploitable in the future to bridge some potential gaps due to lack of skills of the current model.

The KG extraction approach used in this work is based on the idea of leveraging the ability of LLMs to recognize and extract grammar patterns present in the text. Specifically, the grammar pattern that we want to exploit is based on the idea of spotting three fundamental elements: *subject*, *action*, *object*.

Once these items are extracted from the text, they can also be used to further identify entity attributes or other elements of interest. The three-dimensionality of the extracted information makes the subsequent

mapping into a knowledge graph straightforward.

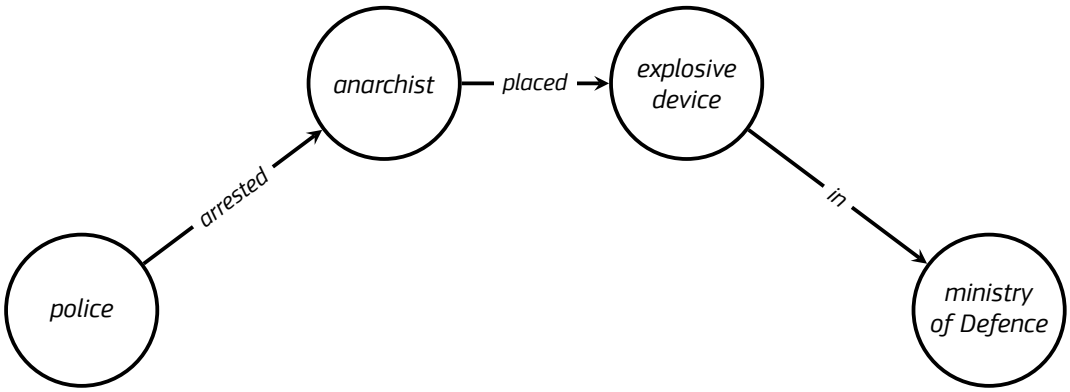
Figure 3.3: Example of text analysis. HYBOLT process the input to produce a Knowledge Graph. Example of analysis output highlighting the triplet set inferred from the input text

```
Input:
An anarchist who placed an improvised explosive device in the Ministry of Defence was
    arrested by the police just minutes later.

Output:
(anarchist, placed, explosive device)
(explosive device, in, ministry of defence)
(police, arrested, anarchist)
```

By applying this approach, we want our solution to receive as input a snippet of text and provide as output a set of triplets summarizing the fundamental meaning of the text while being simple enough to allow the construction of a KG. A practical example of such a process could be the snippet of text in Figure 3.3 provided as input to the system. After the inference phase, we expect the system to be able to recognize and extract patterns such the ones in Figure 3.3 and translate the extracted triplets into the knowledge graph depicted in Figure 3.4.

Figure 3.4: Knowledge Graph built on the set of triplets provided as output by the LLM.

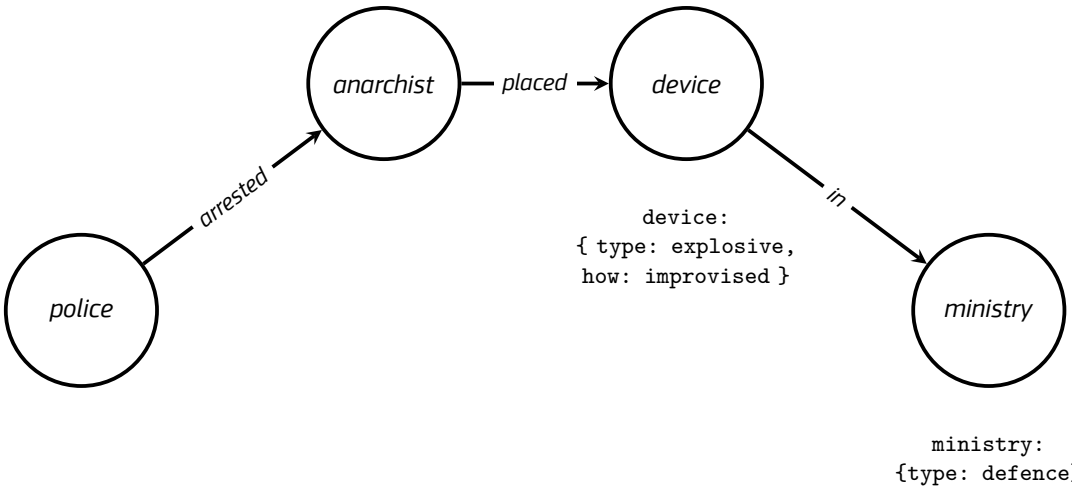


where *anarchist*, *explosive device*, *police*, and *ministry of defence* are the entities, and *placed*, *in*, and *arrested* represent relationships between couple of entities. Comparing the extracted patterns with the original text, it might be possible to note as some information is lost due to the rigid - but needed - constraint of summarizing pieces of information in triplets. Moreover, it is evident that some entity's name might be simplified to obtain a much more readable graph and a set of entity names which are more likely to overlap, thus reducing the overall number of duplicate entities in the KG. For this reason, we want the solution to be able to simplify as much as possible the entity names while, at the same time, enriching the extracted entities with strictly related attributes present in the same piece of text. With this approach, that we will refer to as *atomization* from now on, we aim at building simpler yet more informative and structured entities. Applying this method to the previous example would transform the graph in something similar to what is depicted in Figure 3.5 where entities are named in a much simpler way, and additional information, that might be lost due to the simplification of the name, is linked to the entity as an attribute along with other information present in the text. This approach allows the entities to be simpler, thus the graph to be more readable and informative. All the implementation details of the **triplet extractor** block will be deeply discussed in section 4.

3.3 Performance assessment

The performance of HYBOLT can be evaluated both qualitatively and quantitatively. From a qualitative perspective, one can assess the coherence of the overall graph's structure and examine the semantic quality of the entities and relationships, evaluating whether the system successfully captures the text's meaning and transfers it into the KG.

Figure 3.5: Same Knowledge graph of Figure 3.4 built by exploiting atomization.



For quantitative evaluation, HYBOLT's performance can be assessed by measuring its ability to correctly extract specific information without hallucinating or omitting relevant entities. This evaluation can be structured around a set of downstream tasks that serve as practical examples of how the system might be used in an operational setting. Two such tasks, detailed further in Section 5, include extracting the location of events described in the news and determining the number of individuals affected by these events. In both cases, the system's performance is measured by the percentage of correctly extracted information across a set of news articles (Equation 3.1). For the location extraction task, the analysis can be further segmented by country to identify performance variations across regions and investigate their causes.

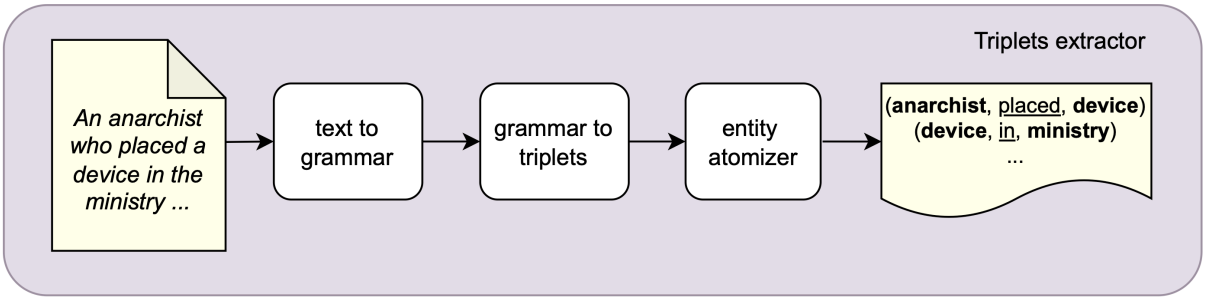
$$\text{accuracy} = \frac{\#\text{correct extractions}}{\#\text{tot news}} \tag{3.1}$$

It is important to note that HYBOLT is task-agnostic, meaning it does not explicitly perform the extraction of specific information from the input text but instead focuses on generating *subject-action-object* triplets. This design choice aims to mitigate hallucination phenomena. Furthermore, the dataset used to evaluate HYBOLT is derived from the European Media Monitor (EMM) platform and comprises news articles about terrorist events collected from various online sources. To comply with copyright regulations, each article includes only a brief portion of text from the beginning of the webpage. As a result, these snippets of text may not consistently contain the specific information required for our downstream tasks. This limitation must be taken into account during the evaluation process. These considerations will be deeply examined in Section 5: Experiments and Results.

4 Implementation strategies

As described in Section 3.2, the primary goal of the ingestion pipeline is to construct a simple yet informative knowledge graph from a given set of news articles, see Figure 3.1. The core component enabling this functionality is the triplet extractor. Its objective is to leverage basic LLM capabilities to minimize hallucinations while extracting meaningful triplets from the input text. Our triplet extraction mechanism operates on the premise that most of the relevant information can be effectively summarized in triplets consisting of a *subject*, an *action*, and an *object*. This assumption aligns well with the Hybrid Threat domain, as defined by the Center of Excellence for Countering Hybrid Threats (Hybrid CoE) (Hybrid-CoE, 2022). Specifically, HT refers to actions planned and executed with harmful intent to destabilize a target (e.g., state, institution, etc.) by using various means. Therefore, this work represents an effort to design a Knowledge Graph creation strategy that is tailored to the HT domain. Our approach is encapsulated in the triplet extractor, which is composed of the three main building blocks illustrated in Figure 4.1. The entire pipeline takes unstructured text as input and generates a set of triplets as output, upon which the system can construct a KG.

Figure 4.1: Flowchart of the main blocks composing the triplets extractor.



Beginning with the *text to grammar* block, its primary purpose is to extract knowledge from unstructured text by leveraging grammar patterns embedded within the data, as its name suggests. Specifically, this block identifies and extracts *subject*, *action*, and *object* elements from the raw text using a few-shot prompting strategy. We provide the LLM with a prompt that requests the extraction of all detectable subjects, actions, and objects from the text, along with a few examples, such as the one shown in Figure 4.2. Examples in the prompt give instructions to the LLM and help to clarify the pattern the LLM should look at. This entire prompt is concatenated with the raw text to be analyzed, and the model outputs a string containing triplets structured in the same format shown in the examples within the prompt. The output string is then passed to the *grammar to triplets* block. This block does not use any LLM capabilities but is instead designed to convert the string into a list of tuples, with each tuple containing a triplet previously extracted by the *text to grammar* block. With the output produced by this block, it becomes possible to construct a KG that summarizes the analyzed text, creating a structure like the one shown in Figure 3.4.

Figure 4.2: Example of the few shot strategy implemented for the extraction of grammar entities from raw text.

```
<example>
Rob took the ball and played volley
<answer>
<SUBJECT>Rob <ACTION>take <OBJECT>ball
<SUBJECT>Rob <ACTION>play <OBJECT>volley
```

However, as the goal of the triplet extractor is to simplify the KG as much as possible, the list of tuples generated by the *grammar to triplets* block is further processed by the *entity atomizer*. This block is responsible for the atomization process described in Section 3.2. It processes all triplets in the list and, leveraging LLM capabilities, deconstructs long and informative entity names (either subject or object) into smaller components, identifying the primary name of the entity while assigning the remaining details as attributes. The result is the creation of a simplified entity enriched with additional information as attributes. This process is achieved by

providing the LLM with instructions aimed at decomposing entity names, followed by a few examples, such as the one shown in Figure 4.3, to implement a few-shot strategy like the *text to grammar* block.

Figure 4.3: Example of the few shot strategy implemented for the atomization of entities extracted from raw text.

```
<example>
Defence ministry
<answer>
{
 "ministry": {
    "type":"defence"
  }
 }
```

After this step, the *entity atomizer* compares the list of tuples resulting from the atomization process with the original list of tuples to ensure consistency. In the same raw text provided to the ingestion pipeline, it is indeed possible for the same entity to appear multiple times, as illustrated in Figure 4.2, where *Rob* is present in both triplets. To prevent the atomization process from converting what was originally recognized as the same entity into different entities with similar names, the system compares the original and atomized triplets for consistency. If inconsistencies are detected, it enforces consistency by assigning the same name to all occurrences of the same entity. This ensures consistent entity naming at least within a single text and increases the likelihood of achieving consistency across multiple texts.
To achieve consistency across multiple texts and build a progressively coherent KB, different strategies may be considered. A *high-resolution* strategy would, for instance, involve comparing text-derived entities with existing KB entities each time a new news-related gaph is generated. This can be approached in several ways: by directly matching atomized entities from the text against the KB, bypassing earlier steps of text-level entity unification; or by unifying entities at the text level first and then validating the unified entities against the KB to ensure alignment. This approach solves potential conflicts or mismatches early on, reducing computational overhead of the integration of new data into the main KB. A *low-resolution* strategy, in contrast, follows a similar validation process but is applied periodically or triggered by specific conditions (e.g., after accumulating a certain number of new entities). Rather than validating entities individually as they are added, this approach consolidates and reviews them in batches. If these sets of entities are homogeneous, for example they are all about a specific theme, or derived from the same sources, the integration process can give more harmonized results enhancing the quality of resulting KGs. The choice between these strategies depends on system requirements and user preferences and they represent a quality/performance trade-off. In scenarios where having an up-to-date KB is critical, accepting higher latency during updates may be justifiable. Conversely, in systems where peak performance is essential during usage, scheduling KB updates during low-traffic periods might be more appropriate.
The outcome of this entire process is intended to be a simple yet informative knowledge graph that can be integrated with the querying pipeline for further analysis of the extracted KB. The proposed solution emphasizes the concept of utilizing basic LLM capabilities, such as grammar structure, to extract knowledge from raw text within the HT domain.
The set of prompts used by the *text to grammar* and *entity atomizer* blocks to reach this objective are depicted in Figure 4.4 and 4.5 respectively. As mentioned before, both prompts employ a few-shot learning strategy, where an initial text explains the task, followed by examples of input-output pairs marked with example-answer tags. For the *text-to-grammar* block, just a couple of examples were sufficient to demonstrate the task and guide the LLM. However, for the *entity atomizer*, four examples were progressively added after a series of experiments. This need for more examples in the *entity atomizer* task can be attributed to the different complexities of the two tasks compared to LLM capabilities. For a smaller model like *nous-hermes-2-mixtral-8x7b-dpo*, tasks such as identifying grammar patterns in the text - as required in the *text to grammar* task - are likely easier to manage than the more intricate *entity atomization*. The latter task involves more than simply breaking down an entity's name. It requires the LLM to interpret each component of the name, determine which word to use as the new name, and decide how to label the attributes that capture the remaining parts of the original name. This process involves multiple steps and requires a certain level of semantic skills. To reduce the number of examples needed in the prompt, a potential strategy could have been a step-by-step, chain-of-thought approach. This method would break down the entity atomization process into

Figure 4.4: Prompt provided to the text to grammar processing block to produce a string of triplets subject,action,object starting from unstructured text.

```
As a grammar expert find all the subjects, the actions performed by
subjects, and objects in the sentences.Use the format <SUBJECT>
<ACTION> <OBJECT> one per output line. Use a few word as possible.
The output has to be just the triplets, no other text.

        <example>
        Rob took the ball and played volley
        <answer>
        <SUBJECT>Rob <ACTION>take <OBJECT>ball
        <SUBJECT>Rob <ACTION>play <OBJECT>volley
        <example>
        Terrorist attacked the city and killed many people
        <answer>
        <SUBJECT>Terrorist <ACTION>attacked <OBJECT>city
        <SUBJECT>Terrorist <ACTION>kill <OBJECT>people
        <example>
        {text_to_analyze}
        <answer>
```

smaller, more manageable tasks, each requiring less complex reasoning overall.

Figure 4.5: Prompt provided to the entity atomizer processing block to simplify the extracted entities into JSON-like structures containing attributes.

```
Given the provided entity, if needed, split it in "name" and "properties".
Do it only if strictly needed, otherwise leave the name as it is. Return
Just the output as JSON with no additional text. If some properties are
boolean, return them between quotes as they were strings. For instance:
          <example>
          Defence ministry
          <answer>
          {"ministry": {"type":"defence"}}
          <example>
          improvised explosive device
          <answer>
          {"device": {"type":"explosive",
                     "how":"improvised"}}
          <example>
          Afghanistan's national intelligence agency
          <answer>
          {"agency": {"type":"intelligence",
                     "where":"Afghanistan"}}
          <example>
          approximate time
          <answer>
          {"time": {"approximate": "True"}}
          <example>
          {entity_to_atomize}
          <answer>
```

5 Experiments and results

This section presents some results obtained by applying the ingestion pipeline described in Section 4. The dataset used for these results is sourced from the EMM platform and consists of news articles related to terrorist events scraped around the web. Each news snippet contains only a few words from the beginning of the webpage to ensure compliance with copyright regulations. Furthermore, the scraped data did not undergo any cleaning process to correct issues such as missing punctuation or incomplete sentences (e.g., where the word limit cuts off a sentence).

While this adds complexity to the overall task, making it more challenging for the ingestion pipeline to translate the ingested content into a coherent knowledge graph (KG), it also simulates more realistic conditions. For example, web scrapers built for specific news sites may fail when the site's structure changes, or new websites may produce imperfect extractions due to limited integration with the scraping engine. By setting this research in a more difficult scenario, we aim to obtain results that closely reflect real-world operational conditions.

The overall qualitative outcome of the results produced by the ingestion pipeline demonstrates the system's ability to effectively transform unstructured text into a descriptive KG. Entities and relationships derived from the same news article are often semantically interconnected, highlighting the system's tendency to avoid simplifying complex, entangled relationships into a single *overall* triplet. Instead, such relationships are typically represented as a series of interconnected sub-triplets. Moreover, the results show an overall good ability of the system to encapsulate quantitative data into entities when they are clearly defined (e.g., *10 affected individuals* rather than *at least 10 affected individuals)*. However, challenges arise when certain entities are insufficiently described, either due to the writing style or premature text truncation. In these cases, the resulting triplets may become completely disconnected from the primary news-related graph. This poses a significant issue, particularly when aiming for the automated creation of a global KG embodying information from a vast collection of news texts. Under such circumstances, it becomes difficult for analysts or algorithms to link these isolated triplets to the main graph, complicating efforts to provide a comprehensive view of the interaction of these entities within the broader HT structured picture. For instance, Figure 5.1 depicts a KG built by the ingestion pipeline after processing the following news:

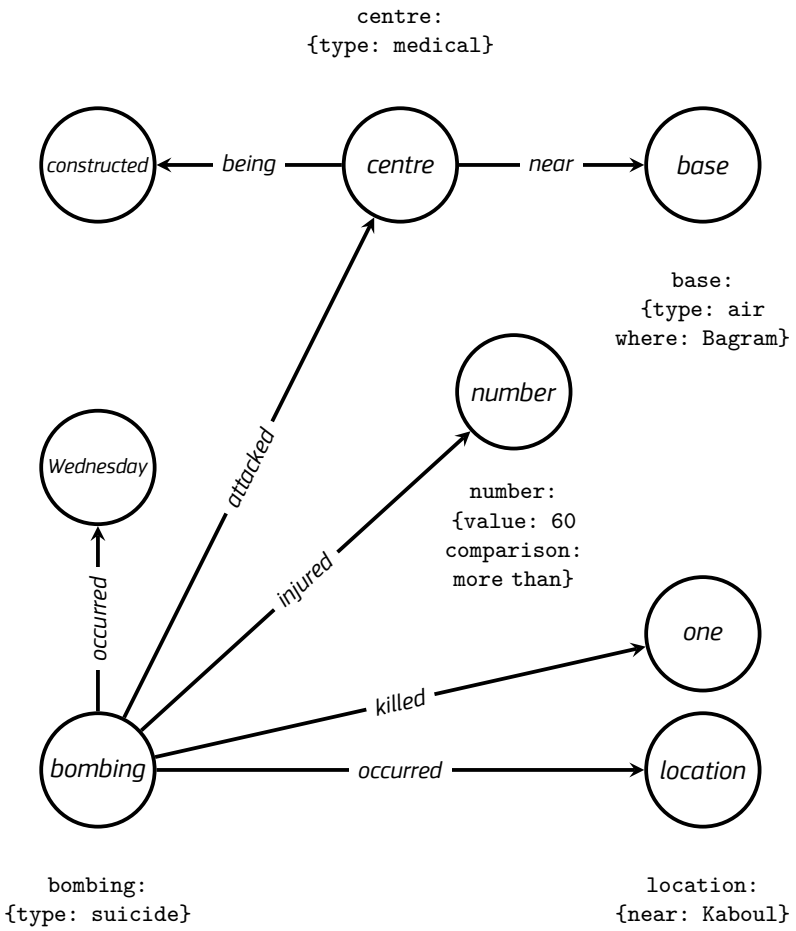
A powerful suicide bombing on Wednesday killed one and injured more than 60 at an under-construction medical centre near Bagram Air Base north of Kabul. No US or coalition personnel were reported among the casualties at the main American military facility in Afghanistan, but the medical centre being....

It is important to note that the suspension points at the end of the news articles, as previously mentioned, are an actual part of the text. They indicate that the news continues, but additional content could not be scraped due to length limitations. Consequently, the ingestion pipeline must account for this source of uncertainty.

From Figure 5.1, it is possible to note some interesting behaviors as well as some issues. One positive outcome is that, in this case, the system recognizes where the event happened and assigns this information to the entity *location*. Another positive feature of the produced knowledge graph is the chain of entities formed from *bombing* to *base* and passing through the entity *centre*. In this case, not only the system could extract the infrastructure where the attack took place but was also able to specify the type of infrastructure (i.e., medical) thanks to the atomization process, and was able to associate it to a fuzzy location indicator (i.e., near the air base in Bagram). Moreover, the produced graph also captures well the quantitative features of the news by two relationships connected to the entity *bombing*, namely *injured* and *killed*. These two relationships allow the graph to correctly express the amount of injured and killed people during the event. Related to that, a negative aspect that can be grasped from Figure 5.1 is that, even if the numbers reported are correct, they do not follow a familiar format. Injured people are represented with the entity *number* which is not explicative, since the *injured* relationship is related to people. A better entity naming could have been *people*. On the same line, also the entity named *one* is not ideal since it does not specify to what that one is referring. Also in this case, a better structure could come from giving the entity the name *people* and specifying *1* in the value attribute of the entity.

Furthermore, Figure 5.2 displays additional triplets generated from the same news article. Unlike those in Figure 5.1, these triplets are not connected to the main graph. While this does not categorically indicate an error, since snippets of news articles can contain unrelated information, in this case some connections are missing. For example, the *American military facility in Afghanistan* is mentioned in the text processed by the ingestion pipeline, yet the first triplet in Figure 5.2 does not clarify its role. Additionally, the second triplet links the entities *personnel* and *casualties*, which is informative. However, it could have been better integrated into

Figure 5.1: First example of graph generated from the ingestion pipeline.

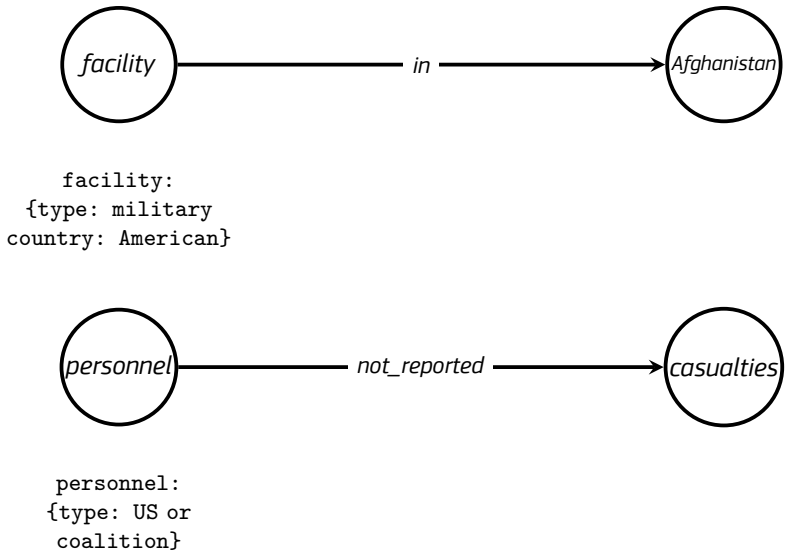


the main graph to indicate that no casualties were reported for this specific attack. Effective integration of such information is essential when combining various news-related graphs into a global knowledge graph, as it ensures a comprehensive and coherent representation of entity interactions.
Another example is depicted in the KG of Figure 5.3 obtained after analyzing the following news:

STORY: Two Russian embassy staff in Kabul were among six people killed when a suicide bomber detonated explosives near the entrance of the embassy, in a blast that injured at least 10 others, the Russian Foreign Ministry and Afghan officials said on Monday.

In this case, the resulting graph demonstrates the system's capability to model the quantitative aspects of the news. By comparing the text snippet with Figure 5.3, it is evident that the number of injured individuals has been correctly linked to the entity *others*, and that the total number of victims and those specifically from the Russian embassy have been accurately associated with the entities *people* and *staff*, respectively. This confirms the system's ability to handle numerical information, as observed in the previous example. However, there is a discrepancy in how the attribute *minimum* is applied to the entity *others*. In Figure 5.1, the entity *number* was linked to two attributes: one representing the value and the other the comparative term. In this case, by contrast, the term *minimum* is used as the descriptive attribute directly associated with the entity *others*, creating inconsistency. Regarding the group of entities *ministry*, *day*, and *officials*, which appear disconnected from the main graph, they might initially seem irrelevant. However, an analysis of the text reveals that all the information available in that subsection of the provided snippet has been successfully captured. The perceived lack of integration between this group and the main graph is likely a consequence of the limited length of the news provided.
For the quantitative evaluation, we conducted an experiment using a stratified random sample of 100 news articles from four distinct locations. The goal was to obtain insights into the system's performance on two downstream tasks: *location extraction* and *extraction of affected individual's quantities*. Before analyzing the results, it is important to note some points. First, the system is not explicitly required to do any of these tasks.

Figure 5.2: Additional set of triplets of the graph in Figure 5.1.



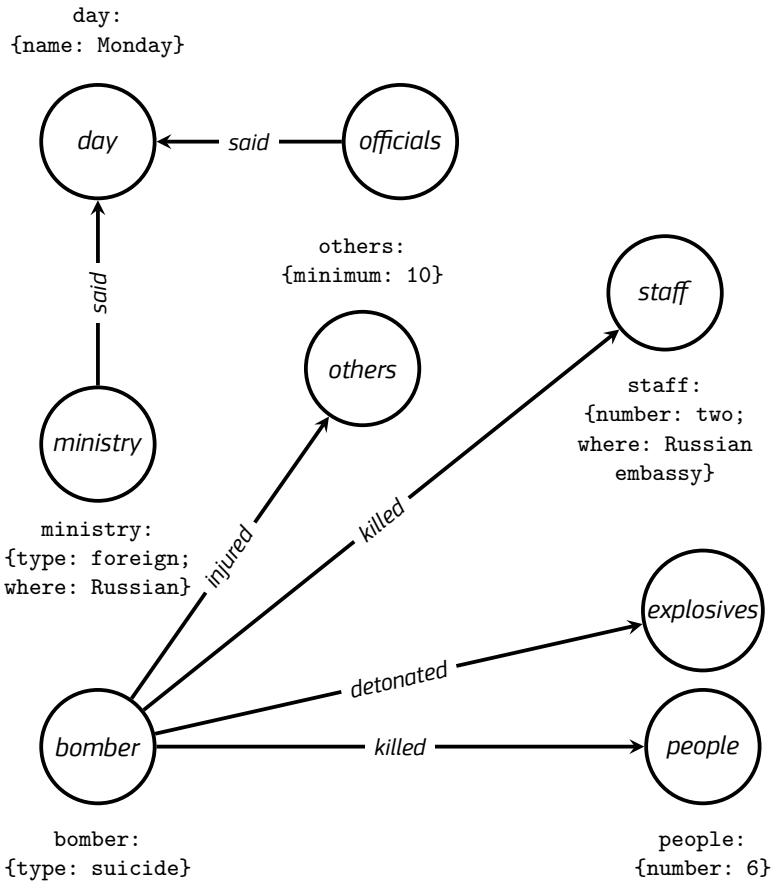
Thus, this analysis primarily assesses its ability to recognize that locations and affected individuals' quantities are key features relevant to the HT domain. Second, the evaluation sample size is limited due to the manual labelling and interpretation required for the assessment. The ingestion pipeline, being a proof-of-concept at this stage, does not yet support automated result evaluation. As highlighted in the preceding qualitative discussion, entities can still appear with slightly varying names and attribute structures, complicating automatic analysis. Additionally, the EMM dataset consists of short news snippets, which are limited in size. The 100 samples were (stratified) randomly selected, meaning not all are guaranteed to mention a location or the number of affected individuals. This endogenous randomness has been kept for two main reasons. A manual selection of news articles could introduce bias, for example, by favoring samples that clearly specify locations and affected individuals, therefore not allowing to reflect the system's potential effectiveness in real-world scenarios. Moreover, the absence of such information in some samples provides an opportunity to test whether the system tends to hallucinate by generating details not in the text. For instance, in news about a terrorist attack, the LLM might infer and add the number of affected individuals even if such information is not explicitly stated in the input text.
Starting from the *location extraction*, the evaluation set comprises samples equally distributed across 4 locations: United States, United Kingdom, Ukraine, and Afghanistan. Across the full set of selected news, 79% mentioned the location of the event, and the remaining 21% did not. In the section not stating the location no hallucination phenomena have been observed. Concerning the set of news reporting the location, in 49% of the cases the location has been correctly extracted, in 49% not extracted and in 2% the extracted location was not the correct one. The overall branching is depicted in Figure 5.4.
It is worth mentioning that also in the case of wrong extraction, no hallucination phenomena have been observed. The wrong extraction was most probably related to the presence of other locations in the same text. For instance, the news

ISLAMABAD: A late night roadside bombing in eastern Afghanistan struck a vehicle carrying members of the Pakistani Taliban group, killing a senior leader and three other militants, several Pakistani officials and militant figures said Monday. No one immediately claimed responsibility for the Sunday....

produced just one entity containing the name of a location, which is the following:

```
{
"officials":
  {
  "number": "several",
  "where": "Pakistan"
  }
}
```

Figure 5.3: Second example of graph generated from the ingestion pipeline.



Even if for a human reader this clearly refers to the nationality of the officials involved, it is important to keep in mind that a potential analyst is not required to have access to the original news, and without it, just by looking at the previous entity, one would most likely tend to think that the location of the attack was Pakistan. Moreover, the generated attribute name *where* is misleading. A better name would have been, for instance, *from* since it refers to the nationality of a group of people. Concerning the portion of news in which the system did not extract any location even though present in the text (purple branch in Figure 5.4) can be seen as a way of confirming the low tendency of the system to hallucinate. This trait of the system is derived from the absence of an explicit request to extract the location from the text. The results show that, if not explicitly asked, the system will extract the location information just if it is relevant and clearly defined within the provided piece of text.

By segmenting the obtained results by location, as shown in Figure 5.5, we observe that for news related to the United States, in 56% of the cases, the location was not assessed by the system. Upon analyzing the data, this issue appears strongly related to the tendency of US-related news to reference locations using counties or districts rather than explicitly mentioning the state or nation. Even when the nation's name is mentioned, the text often associates the event with multiple geographical levels across different parts of the text (e.g., First part: *shooting Wednesday night in the Lower Valley*; Second part: *The teenager was hit [...] in the 100 block of Edith Drive near Alameda Avenue and Delta Drive in the Ascarate neighborhood*). The results show that within this 56% subset, no sample directly referenced a specific and clear location. Instead, the text frequently mentioned various locations on multiple levels, often omitting the nation's name. While this might seem only partially relevant, it can introduce several sources of uncertainty, such as confusion due to homonymous locations (e.g., Athens, Georgia; Birmingham, Alabama). For instance, it is more common to encounter text like *"Bergen County man was gunned down in Jersey City"* rather than *"Man was gunned down in New Jersey, USA."* This leads to increased uncertainty in identifying the locations mentioned in the news. Similar statistics apply to the United Kingdom and Afghanistan, where the ratio for *correctly/not extracted* locations (●/●) is also low. In these cases, respectively 60% of the 40% for Afghanistan and 100% of the 36% for the United Kingdom were not reporting a clear and univocal statement of the location where the event happened.

Figure 5.4: Sankey diagram of the location assessment. Evaluation set indicates the total number of news used for the assessment. Location● and No location ● indicate respectively the set of news where the location was or was not mentioned in the text. Wrong●, Right●, and No Extraction● refer to the sets of news respectively being, wrongly, rightly, or not assessed by the system

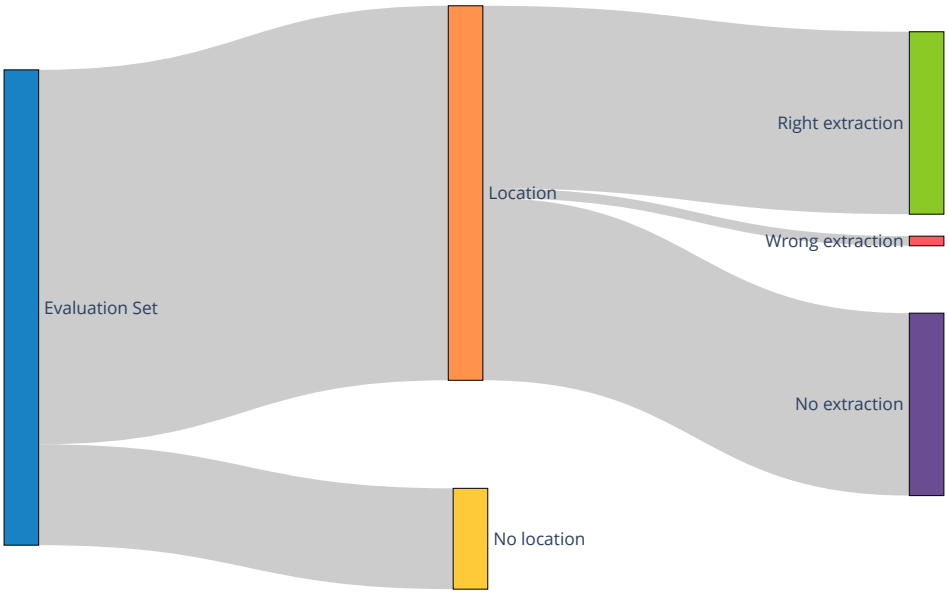
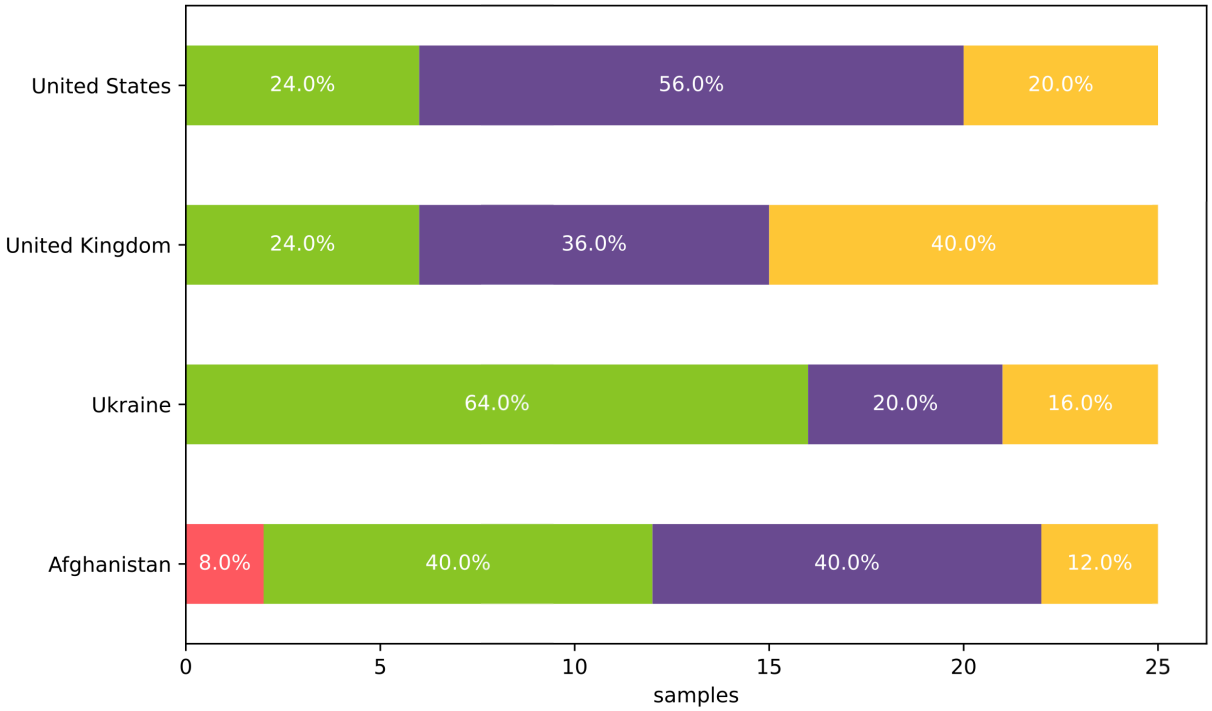


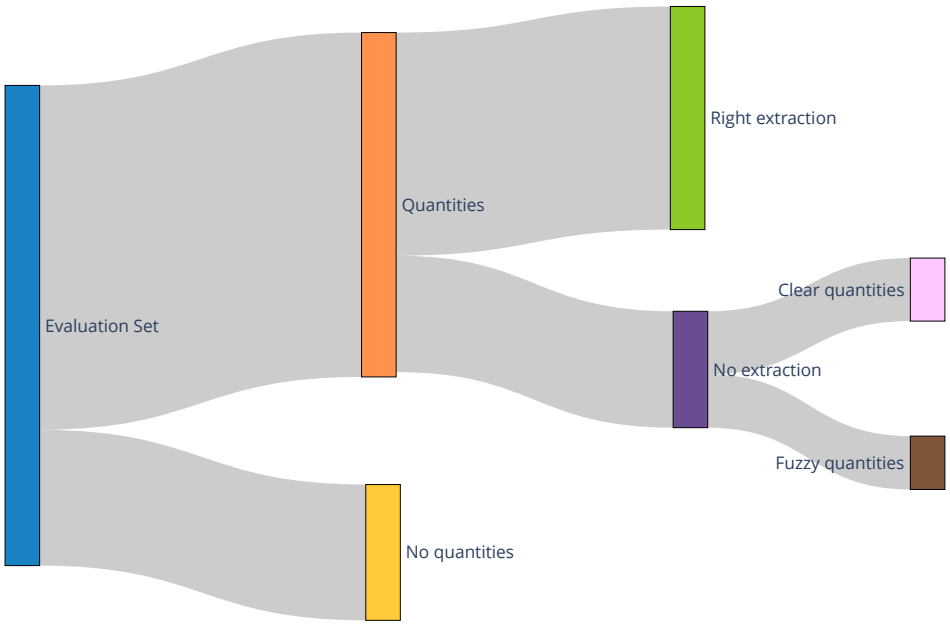
Figure 5.5: Segmentation by location of the percentages of Wrong●, Right●, No Extraction●, and No Location● news. The color code is the same as Figure 5.4.



With a similar but complementary assumption, we can explain the 64% of correct location extraction for Ukraine. Much of the news relates to the ongoing conflict that began in February 2022, which has impacted multiple regions within the country. Given Ukraine's smaller size compared to, for example, the United States, news reports are more likely to explicitly mention "Ukraine". This increases the system's confidence in identifying the correct location and leads to a higher *correctly/not extracted* location rate. The only location with recorded errors is Afghanistan, where inaccuracies are due to mention of the nationality of military or paramilitary forces involved in the event.

For the analysis on the *extraction of affected individual's quantities* downstream task, the same evaluation set used for the *location extraction* was used. Within the dataset, 71% of the news articles explicitly mentioned the quantity of affected individuals, while the remaining 29% did not. In the latter group, no hallucination phenomena were observed, meaning the system did not generate non-reported quantities. Among the news articles reporting quantities, the system successfully extracted the correct numbers in 65% of the cases, while in the remaining 35%, no extraction was performed. No quantity extraction has been observed to be wrong. The overall branching is visualized in Figure 5.6. Interestingly, the 35% of samples without any quantity extraction can be further categorized into two groups. The first, *Clear Quantities*, accounts for 54% of these cases and includes samples where the quantities are explicitly stated. The second, *Fuzzy Quantities*, comprises the remaining 46% and includes samples where the quantities are prefixed with the term "*at least*", making them less certain. As a result, half of the *No Extraction* cases involve ambiguously defined quantities, which prevents the model from identifying and extracting the number of affected individuals. This behavior is particularly interesting since all samples within the *Fuzzy quantity* category, shown in Figure 5.6, shared the exact same prefix: "at least". Conversely, it is also worth mentioning that a small proportion of the *Right extraction* set included fuzzy quantities. Interestingly, the graph example in Figure 5.3 is based on a piece of news containing the sentence "at least 10". Despite that, the system successfully extracted the number and interpreted the fuzzy prefix correctly. The logical conclusion is that fuzzy prefixes such as "at least" increase the system's uncertainty regarding the targeted quantity, reducing the likelihood of extracting the information as a standalone entity.

Figure 5.6: Sankey diagram of the affected individual's quantity assessment. Evaluation set indicates the total number of news used for the assessment. Quantities ● and No quantities ● indicate respectively the set of news where the quantities of affected individuals were or were not mentioned in the text. Right ● and No extraction ● refer to the sets of news respectively being rightly, or not assessed by the system. Clear ● and Fuzzy quantities ● refer to the set of news respectively reporting a clear quantity or a quantity preceded by the prefix “at least”.



6 Conclusions and future work

This work serves as a proof-of-concept and an exploration of the feasibility of developing a system capable of converting unstructured text into a structured knowledge graph tailored to the Hybrid Threats domain. The aim is to enable the extraction of meaningful and actionable intelligence from open-source data, such as news articles, to enhance the efficiency and effectiveness of analysts who frequently work with diverse data sources that require integration. We chose two downstream tasks that are easy to understand and verify but the ideas presented are general and can be applied to many other tasks.

The combination of qualitative and quantitative analyses in Section 5 highlights the proposed system's strong information extraction capabilities and its resilience against hallucination phenomena. A comparison of results and methodology suggests that this success can be attributed to the segmentation of the transformation process from unstructured text to structured output into distinct sub-steps, including grammar analysis. This approach guides the LLM through the process rather than requiring it to handle the entire transformation independently. Also, the decision not to explicitly request the extraction of specific items helps reduce hallucinations. As demonstrated in the two downstream tasks discussed in Section 5, there were no explicit instructions to extract locations or quantities of affected individuals. The system did not extract any locations or quantities that were not present in the provided text, further showing its resilience toward hallucination.

Conversely, still concerning one of the downstream tasks, the system seems to struggle with the extraction of not well and uniquely defined locations. In multiple cases, it was not possible to extract locations when the text was associating the occurrence of the explained event to multiple geographical levels across distinct parts of the news. The association of the event to the correct location with different geographical granularity seems to *confuse* the LLM resulting in the information not being extracted. A potential solution to address these challenges could involve integrating geocoding or Named Entity Recognition (NER) modules into the system. These enhancements would enable a more systematic approach to handling location information (Bosso et al., 2023). Before developing ad-hoc solution it is worth exploring more accurately the structure of the prompts that constitute the LLM instructions. An interesting approach can compare the different KGs produced by different prompts to decide the most promising one.

Regarding the second downstream task, as highlighted in both the qualitative and quantitative analyses, there is room for improvement in unifying entities that report quantities of affected individuals. While the system demonstrates a preference for reliability over completeness, highlighted by the absence of incorrect extractions or hallucinations, further refinement is needed to achieve consistent entity unification. The current system struggles to maintain a coherent nomenclature for entities representing the same concept across different texts. This inconsistency poses significant challenges for constructing a global knowledge graph by integrating overlapping information present in individual news-related graphs.

In the future to overcome this KG integration issue we plan to complement these first steps with other machine learning techniques. A first attempt will be the use of similarity measure based on semantic embeddings to help the mapping of similar terms and concept in a uniform way across the same text. With the help of semantic vector databases and the same semantic embeddings, it will be possible to explore the KG already acquired to find appropriate concept to link to. In this way we will implement the idea of a representation with standard terms, concepts, and patterns. A crucial step is the production of a pre-defined core KG representing some main concept so that the whole KG will be built in the user defined style. If this core KG also includes typical patterns, it will be possible to use these to express common occurrences. For example, to express casualties in an attack the KG will always use the same concepts and relationships. This is important because for downstream tasks we need to know how to correctly express our queries over the KG being sure of capturing all the interesting patterns. The Hybrid Threat CORE model presented in (Jungwirth R et al., 2023) is an ideal starting point for the construction of a core KG. In the CORE model HT are presented in a rational way that naturally resemble the way subject-relationship-object triple at the base of the KG are presented. This similitude in the structures will ease the core KG construction work, that at least at the beginning, will be mostly a manual effort. At the same time if the core KG does not contain the term and concepts needed, we need a way to expand it. The proposed HYBOLT framework already has enough flexibility to accommodate this task. However, several parameters will be necessary to establish the right trade-off between accurate representation of the text and standardization in the KG. Naturally, the more the KG is accurate the less the representation is standard. This is also an important work stream needed to maximize the usefulness of this kind of system also considering possible specific downstream tasks. This work represents a reasonable starting point for future research activities in ontology learning applied to the hybrid threats domain.

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List of abbreviations and definitions

JRC	Joint Research Centre of the European Commission
HT	Hybrid Threats
CORE	Comprehensive Resilience Ecosystem
OSINT	Open Source INTelligence
AI	Artificial Intelligence
NLP	Natural Language Processing
LLM	Large Language Model
DL	Deep Learning
HPC	High-Performance Computing
GPU	Graphical Processing Unit
TPU	Tensor Processing Unit
LPU	Language Processing Unit
KG	Knowledge Graph
OL	Ontology Learning
CQ	Competency Question
RAG	Retrieval Augmented Generation
HYBOLT	HYbrid-threat Based Ontology Learning Tool
RDF	Resource Description Framework
EMM	European Media Monitor
NER	Named Entity Recognition

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